

4.4.18

ai25btech11022 - Narshitha

October 4, 2025

Question:

If the point **A** (x, y), **B** (3, 6) and **C** (-3, 4) are collinear, show that $x - 3y + 15 = 0$

Solution:

As **A**, **B** and **C** are collinear

$$\text{rank}(\mathbf{A} - \mathbf{B} \quad \mathbf{B} - \mathbf{C})^T = 1 \quad (0.1)$$

$$\begin{pmatrix} x-3 & y-6 \\ x+3 & y-4 \end{pmatrix} \quad (0.2)$$

$$R_2 \longrightarrow R_2 - \frac{x+3}{x-3} R_1$$

$$\begin{pmatrix} x-3 & y-6 \\ 0 & y-4 - (y-6) \frac{x+3}{x-3} \end{pmatrix} \quad (0.3)$$

$$\implies y-4 - (y-6) \frac{x+3}{x-3} = 0 \quad (0.4)$$

$$\therefore x - 3y + 15 = 0 \quad (0.5)$$

Collinearity of A, B, and C

